Functions

Local VS Global Variables

Review

Both main function, and addition function has different set of variables.

```
1   #include <iostream>
2   using namespace std;
3
4   int addition(int num1, int num2);
5
6   main(){
7       int number1, number2, result;
8       cout << "Enter First Number: ";
9       cin >> number1;
10      cout << "Enter Second Number: ";
11      cin >> number2;
12      result = addition(number1, number2);
13      cout << "Sum is: " << result;
14  }
15
15  int addition(int num1, int num2)
16  {
17      int sum = num1 + num2;
18      return sum;
    }
```

Function with no parameters

Instead of passing parameters, can we use same parameters?
i.e.,

number1, number2 and result

```
1   #include <iostream>
2   using namespace std;
3
4   int addition(int num1, int num2);
5
6   main(){
7       int number1, number2, result;
8       cout << "Enter First Number: ";
9       cin >> number1;
10      cout << "Enter Second Number: ";
11      cin >> number2;
12      result = addition(number1, number2);
13      cout << "Sum is: " << result;
14  }
15
15  int addition(int num1, int num2)
16  {
17      int sum = num1 + num2;
18      return sum;
    }
```

Function with no parameters

Instead of passing parameters, can we use same parameters?
i.e.,

number1, number2 and result

```
1   #include <iostream>
2   using namespace std;
3
4   int addition();
5
6   main(){
7       int number1, number2, result;
8       cout << "Enter First Number: ";
9       cin >> number1;
10      cout << "Enter Second Number: ";
11      cin >> number2;
12      result = addition();
13      cout << "Sum is: " << result;
14  }
15
16  int addition()
17  {
18      result = number1 + number2;
19      return result;
20  }
```

Error

```
example.cpp: In function 'int addition()':
example.cpp:18:5: error: 'result' was not declared in this scope
   18 |     result = number1 + number2;
      |     ^~~~~~
example.cpp:18:14: error: 'number1' was not declared in this scope
   18 |     result = number1 + number2;
      |              ^~~~~~~
example.cpp:18:24: error: 'number2' was not declared in this scope
   18 |     result = number1 + number2;
      |                        ^~~~~~~
```

```cpp
#include <iostream>
using namespace std;

int addition();

main(){
    int number1, number2, result;
    cout << "Enter First Number: ";
    cin >> number1;
    cout << "Enter Second Number: ";
    cin >> number2;
    result = addition();
    cout << "Sum is: " << result;
}

int addition()
{
    result = number1 + number2;
    return result;
}
```

Local Variables

Variables within a block { } remain accessible only within that block and not outside that block. These are called local variables of block.

```
1   #include <iostream>
2   using namespace std;
3
4   int addition();
5
6   main(){
7       int number1, number2, result;
8       cout << "Enter First Number: ";
9       cin >> number1;
10      cout << "Enter Second Number: ";
11      cin >> number2;
12      result = addition();
13      cout << "Sum is: " << result;
14  }
15
16  int addition()
17  {
18      result = number1 + number2;
        return result;
    }
```

Scope of Variables

The area within a variable can be accessed is called **scope** of the variable.

```
1   #include <iostream>
2   using namespace std;
3
4   int addition();
5
6   main(){
7       int number1, number2, result;
8       cout << "Enter First Number: ";
9       cin >> number1;
10      cout << "Enter Second Number: ";
11      cin >> number2;
12      result = addition();
13      cout << "Sum is: " << result;
14  }
15
16  int addition()
17  {
18      result = number1 + number2;
        return result;
    }
```

Solution: Global Variables

We can Declare Global Variables before the main function.

```
1   #include <iostream>
2   using namespace std;
3
4   void addition();
5   int number1, number2, result;
6   main(){
7       cout << "Enter First Number: ";
8       cin >> number1;
9       cout << "Enter Second Number: ";
10      cin >> number2;
11      addition();
12      cout << "Sum is: " << result;
13  }
14
15  void addition()
16  {
17      result = number1 + number2;
18
    }
```

Solution: Global Variables

We can Declare Global Variables before the main function.

```
C:\C++>c++ example.cpp -o example.exe

C:\C++>example.exe
Enter First Number: 5
Enter Second Number: 9
Sum is: 14
C:\C++>
```

```
1   #include <iostream>
2   using namespace std;
3
4   void addition();
5   int number1, number2, result;
6   main(){
7       cout << "Enter First Number: ";
8       cin >> number1;
9       cout << "Enter Second Number: ";
10      cin >> number2;
11      addition();
12      cout << "Sum is: " << result;
13  }
14
15  void addition()
16  {
17      result = number1 + number2;
18
    }
```

Local Variables

```
#include <iostream>
using namespace std;

int addition(int num1, int num2);

main(){
    int number1, number2, result;
    cout << "Enter First Number: ";
    cin >> number1;
    cout << "Enter Second Number: ";
    cin >> number2;
    result = addition(number1, number2);
    cout << "Sum is: " << result;
}

int addition(int num1, int num2)
{
    int sum = num1 + num2;
    return sum;
}
```

Low Coupled

Which one is better?

Global Variables

```
#include <iostream>
using namespace std;

void addition();
int number1, number2, result;
main(){
    cout << "Enter First Number: ";
    cin >> number1;
    cout << "Enter Second Number: ";
    cin >> number2;
    addition();
    cout << "Sum is: " << result;
}

void addition()
{
    result = number1 + number2;

}
```

High Coupled

Local Vs Global Variables

- Low Coupling is Good and Always Desired.
- In Some Cases, where multiple function need to share the same data we have to declare GLOBAL variables.

Self Assessment

1. What will be the output of the program?

```
#include <iostream>
using namespace std;

/* global variable declaration */
int g = 20;
main()
{
    /* local variable declaration */
    int g = 10;
    cout << "Value of g = " << g;
}
```

Self Assessment

2. What will be the sequence of the output of the program? How many global variables, local variables of main, parameters of sum function and local variables of sum function are there?

```
#include <iostream>
using namespace std;

int a = 20;
int sum(int a, int b);
main ()
{
  int a = 10;
  int b = 20;
  int c = 0;
  cout << "value of a in main() = " << a << endl;
  c = sum( a, b);
  cout << "value of c in main() = " << c << endl;
}
/* function to add two integers */
int sum(int a, int b)
{
   cout << "value of a in sum() = " << a << endl;
   cout << "value of b in sum() = " << b << endl;
   return a + b;
}
```

Self Assessment (With Example)

Blood types are named according to three factors:

1. presence of antigen A
2. presence of antigen B, and
3. presence of Rh factor.

If antigen A is found, the blood type includes the letter "A". If antigen B is found, the blood type includes the letter "B". If both antigens are found, the blood type includes the letter “AB”. If neither antigen A nor antigen B are found, the blood type includes the letter "O".
if the Rh factor is present, the blood type ends with "+"; otherwise, it ends with "-".

Self Assessment

First write antigenChecker function than take two Boolean input that represent either of each antigenA or antigenB is present. This function shall return the antigen type. According to following criteria.

If antigen A is found, the blood type includes the letter "A". If antigen B is found, the blood type includes the letter "B". If both antigens are found, the blood type includes the letter “AB”. If neither antigen A nor antigen B are found, the blood type includes the letter "O".

Self Assessment

First write antigenChecker function than take two Boolean input that represent either of each antigenA or antigenB is present. This function shall return the antigen type. According to following criteria.

What will be the header of function ?

Self Assessment

First write antigenChecker function than take two Boolean input that represent either of each antigenA or antigenB is present. This function shall return the antigen type. According to following criteria.

What will be the header of function ?

```
string antigenChecker(bool antigenA, bool antigenB)
```

Self Assessment

First write antigenChecker function than take two Boolean input that represent either of each antigenA or antigenB is present. This function shall return the antigen type. According to following criteria.

```
string antigenChecker(bool antigenA, bool antigenB)
{
    //What will be the BODY of function ?

}
```

Solution: antigenChecker

Let's write the
string antigenChecker(bool antigenA, bool antigenB)
function first.

```
string antigenChecker(bool antigenA, bool antigenB)
{
    if(antigenA == true && antigenB == false){
        return "A";
    }
    else if(antigenA == false && antigenB == true){
        return "B";
    }
    if(antigenA == true && antigenB == true){
        return "AB";
    }
    else{
        return "O";
    }
}
```

Self Assessment

Now your task is to write a function that decided the rh factor of the blood based on following criteria.

if the Rh factor is present, the blood type ends with "+"; otherwise, it ends with "-".

What will be the function header ?

Self Assessment

Now your task is to write a function that decided the rh factor of the blood based on following criteria.

if the Rh factor is present, the blood type ends with "+"; otherwise, it ends with "-".

What will be the function header ?

```
char rhChecker(bool rH)
```

Self Assessment

Now your task is to write a function that decided the rh factor of the blood based on following criteria.

if the Rh factor is present, the blood type ends with "+"; otherwise, it ends with "-".

```
char rhChecker(bool rH)
{
  // What will be the body of function
}
```

Solution: rhChecker

```
char rhChecker(bool rH)
{
    if(rH == true)
    {
        return '+';
    }
    else
    {
        return '-';
    }
}
```

Self Assessment

Now write a bloodType function that take three argument as follow and print the blood group on the screen

void bloodType(bool antigenA, bool antigenB, bool rH)

This function should use following two functions that we alrready defined.

1. string antigenChecker(bool antigenA, bool antigenB)
2. char rhChecker(bool rH)

Self Assessment

Note:
Take 1 as input from the user if antigen A is present and 0 otherwise. Same is the case for antigen B and rH factor.

Test Cases:

Input	Output
bloodType(1, 1, 0)	AB-
bloodType(1, 0, 1)	A+
bloodType(0, 0, 0)	O-

Solution: bloodType

```
void bloodType(bool antigenA, bool antigenB, bool rH)
{
    string name;
    char sign;
    name = antigenChecker(antigenA, antigenB);
    sign = rhChecker(rH);
    cout << "Blood Type is: " << name << sign;
}
```

Solution: main

Now, Let's write the main function.

```
#include <iostream>
using namespace std;

void bloodType(bool antigenA, bool antigenB, bool rH);
string antigenChecker(bool antigenA, bool antigenB);
char rhChecker(bool rH);

main()
{
    bool a, b, r;
    cout << "is Antigen A present?: ";
    cin >> a;
    cout << "is Antigen B present?: ";
    cin >> b;
    cout << "is rH factor present?: ";
    cin >> r;
    bloodType(a, b, r);
}
```

```
string antigenChecker(bool antigenA, bool antigenB)
{
    if(antigenA == true && antigenB == false){
        return "A";
    }
    else if(antigenA == false && antigenB == true){
        return "B";
    }
    if(antigenA == true && antigenB == true){
        return "AB";
    }
    else{
        return "O";
    }
}
```

```
#include <iostream>
using namespace std;

void bloodType(bool antigenA, bool antigenB, bool rH);
string antigenChecker(bool antigenA, bool antigenB);
char rhChecker(bool rH);

main()
{
    bool a, b, r;
    cout << "is Antigen A present?: ";
    cin >> a;
    cout << "is Antigen B present?: ";
    cin >> b;
    cout << "is rH factor present?: ";
    cin >> r;
    bloodType(a, b, r);
}
```

```
char rhChecker(bool rH)
{
    if(rH == true){
        return '+';
    }
    else{
        return '-';
    }
}
```

```
void bloodType(bool antigenA, bool antigenB, bool rH){
    string name;
    char sign;
    name = antigenChecker(antigenA, antigenB);
    sign = rhChecker(rH);
    cout << "Blood Type is: " << name << sign;
}
```